

# COMSATS UNIVERSITY ISLAMABAD



## PROJECT REPORT

|              |                                              |
|--------------|----------------------------------------------|
| Course       | Business Data Analysis                       |
| Project Idea | Time series data preparation for forecasting |
| Section      | BBD-4A                                       |

## GROUP MEMBERS

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## Dataset description:

### Overview:

This dataset contains weekly sales data of Walmart stores. It shows how much different Walmart stores sold each week, along with some other factors like temperature, fuel price, and unemployment rate, allowing us to analyze trends over time.

- **Total Records:** 6435
- **Total Columns:** 8
- **Time Frequency:** Weekly data

### Columns explanation:

**Store** column represents the store number as multiple stores are included in the dataset, it helps compare performance between different stores

The dataset includes a **date** column in the format DD-MM-YYYY, which shows the week of sales. Because it contains time-based data, it is a time series dataset. This allows us to study how sales change over time.

The most important column is **Weekly\_Sales**, because it tells us how much each store earned in a specific week. This is the main thing we will analyze and later use for forecasting.

the **Holiday\_Flag** shows whether that week had a holiday or not, and usually sales increase during holidays

- 1 = Holiday week
- 0 = Non-holiday week

The **Temperature** column gives the weather conditions, which can also influence shopping behavior such as seasonal trends.

Similarly, **Fuel\_Price**, **CPI (inflation)**, and **Unemployment rate** help us understand the economic conditions during that time.

Overall, the dataset not only shows sales over time but also includes different factors that can affect those sales. This makes it perfect for us to analyze trends, understand patterns, and build forecasting models.

## Code explanation:

- **Step no 1:**

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

first of all we have to import libraries that we need for data analysis. Pandas as pd is used to analyze data especially in table format (like Excel sheets). We use it to read, clean, and perform operations on time series dataset.

matplotlib.pyplot as plt is used to create graphs and charts, that will help us to visualize trends, patterns, and changes in sales over time.

- **Step no 2:**

```
data = pd.read_csv('walmartds.csv')
```

```
data.head()
```

```
data.columns
```

First, we load the dataset from CSV file into Python

then we store the dataset into a variable called data so we can use it later

data.head() will show the first 5 rows of dataset, helps to quickly see what data looks like.

Then columns are used to view the column names

- **Step no 3:**

```
data['Date'] = pd.to_datetime(data['Date'], dayfirst=True)
```

```
data.set_index('Date', inplace=True)
```

Now, we have to change date column into proper format that python understands so we will convert it into datetime. (dayfirst =True) means the date format is day-month-year

then, we'll make date column the index of dataset (inplace =True) means it will not create a new column just updates the current one

- **Step no 4:**

```
monthly_data = data['Weekly_Sales'].resample('ME').sum()
```

```
monthly_data = monthly_data.to_frame()
```

now we are taking only the weekly\_Sales column from dataset and applied resampling on it.

resample is to change the time frequency of the data.

ME means month end, so we are converting weekly sales into monthly sales.

Since there are multiple weeks in a month so sum () will add all weekly sales of that month together

After resampling, the output is in series so we convert it into table i.e DataFrame so it's easier to work with.

- **Step no 5:**

```
monthly_data['Moving_Average'] =
```

```
monthly_data['Weekly_Sales'].rolling(window=3).mean()
```

since our data is now monthly, so rolling(windows = 3) means we'll look at 3 months at a time i.e It take the sales of 3 consecutive months, after first 3 months, we'll move to one month forward and take next 3 months and repeat

then it calculates average of the 3 months in the window, storing that in Moving\_Average

- **Step no 6:**

```
monthly_data['Rolling_Mean'] = monthly_data['Weekly_Sales'].rolling(window=3).mean()
```

```
monthly_data['Rolling_Std'] = monthly_data['Weekly_Sales'].rolling(window=3).std()
```

rolling is method in Pandas that looks at fixed number of past periods here it is 3 months, it moves one step forward at a time. Here Rolling\_mean shows average sales of last 3 months

and Rolling\_Std is to calculate standard deviation which tells how much sales are moving up and down in that 3 month period. A bigger numbers means sales are more spread out and smaller mean sales are stable.

- Step no 7:

```
monthly_data[['Weekly_Sales', 'Moving_Average']].plot()
```

```
plt.title("Sales Trends")
```

```
plt.show()
```

here we are picking two columns from our monthly data

**Weekly\_Sales** is the total sales each month

**Moving\_Average** is the smooth average of the last 3 months

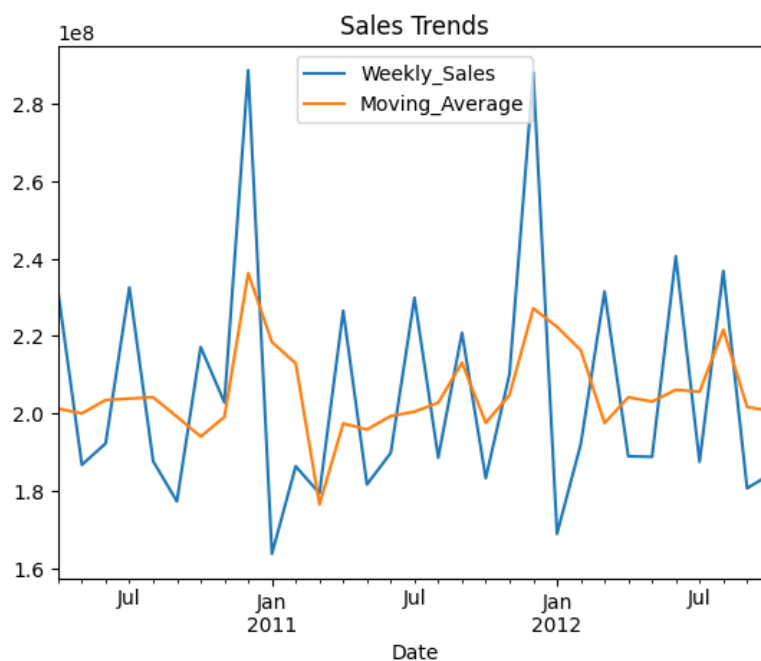
Then we plot a line chart of these two columns, The x-axis is Date (months), the y-axis is sales.

We get two lines:

- One shows the actual monthly sales (wiggly line)
- One shows the moving average (smoother line)

The `plt.title("Sales Trends")` is to add the title to the chart

And then we use `plt.show()` to displays the chart on your screen



- **Step no 8:**

**monthly\_data.head()**

monthly\_data now has:

Weekly\_Sales: total sales each month

Moving\_Average: 3-month moving average using rolling (specifically mean)

Rolling\_Mean: can calculate anything over a sliding window: mean, sum, standard deviation, etc

Rolling\_Std: how much sales are fluctuating

.head()

This shows the first 5 rows of the table.

It's useful to quickly see what your data looks like without printing the whole dataset.

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